a) $f(x) = 5x^{(3-1)} + \cancel{2}$ - subtract 1 from exponent

$f'(x) = 3x^2$ - remove the constant

b) $\frac{-1}{x^6}$ - remove fraction

$= -x^{-6}$

$f'(x) = 6x^{-7}$ - change to derivative

$= \frac{6}{x^7}$ - remove negative

c) $y = (2x)^4$ - Plug in exponant

$y = 16x^{(4-1)}$ - find deriviative

$y' = 64x^3$

d) $g(x) = (3x+4)(2x-1)$ - FOIL and simplify

$= 6x^2 + 5x - 4$

$g'(x) = 12x + 5$ - Find derivative

e) $-\sqrt[3]{x^4}$ - change to fraction

$\frac{dy}{dx} = -x^{4/3}$ - find derivative

$= -\frac{4}{3}x^{1/3}$

f) $f(x) = x(x^2 - 5)$ - Foil

$= x^3 - 5x$ - Find

$\frac{df(x)}{dx} = 3x^2 - 5$ deriviative

g) $\frac{5x^6 + 12x^5}{(3x^2)^2}$ - Simplify denom

$= \frac{5x^6 + 12x^5}{9x^4}$

$= \frac{5x^{\cancel{6}\,2}}{\cancel{9}} + \frac{\overset{\div 3}{12}x^{\cancel{5}\,1}}{\underset{\div 3}{\cancel{9}}\cancel{x^4}}$ - cancel out what you can.

$= \frac{5x^2}{9} + \frac{4x^1}{3}$ - simplify

$y' = \frac{10x^1}{9} + \frac{4^0}{3}$ - find deriviative

Calculate the slope of the tangent line of $y = -x^3 + 2.5x^2 - 3x + 1$ at $x = 2$.

$y' = -3x^2 + 5x - 3$ - Find the derivative

$= -3(2)^2 + 5(2) - 3$ - Plug in $x = 2$

$= -12 + 10 - 3$

$= -5$ - simplify

Determine the equation of the tangent at the point $(2, -3)$.

$y = -3 \quad x = 2$

$m = -5 \quad b = ?$

$y = mx + b$ - Plug in what you know

$(-3) = (-5)(2) + b$

$7 = b$ - find b

$y = -5x + 7$ - plug into $y = mx + b$ to make equation.

The height of a diver above water level is $h(t) = -4.9t^2 + 3t + 20$

a) Calculate his height at 0.5s

Use $h(t)$ $\quad h(0.5) \doteq 20.3$

$\therefore$ It is approx 20.3 m.

b) Calculate IRC at 2.7s.

Use derivative $\rightarrow h'(t) = -9.8t + 3$

$= -23.5$

$\therefore$ IRC @ 2.7s is $-23.5 \frac{m}{s}$

When does the diver break the surface of the water & w/ what velocity?

$h(t) = -4.9t^2 + 3t + 20$ - use $h(t)$

$0 = -4.9t^2 + 3t + 20$ - set $h(t)$ to zero

- use quadratic formula to solve

$\cancel{t \doteq -1.7}$

$t \geq 0$

$t \doteq 2.3$

$\therefore$ Diver breaks the surface of the water at 2.3 sec.

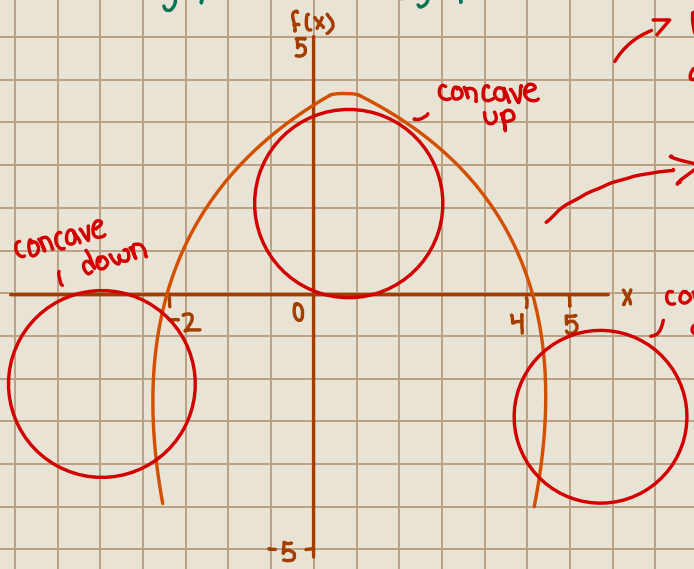
$h'(t) = -9.8t + 3$ - use deriviative

$h'(2.3) = -19.5$ - plug in time

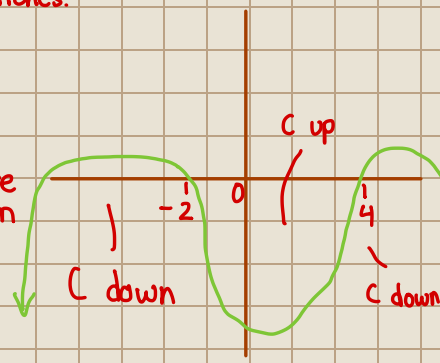
$\therefore$ The diver's velocity is approx $-19.5 \frac{m}{s}$.

Example

Given the graph of $y = f''(x)$, graph $y = f(x)$.



Using the graph, find concavity and which points concavity switches.



- $f''(x)$ is concave down ($f''(x) < 0$) on $(-\infty, -2) \cup (4, \infty)$
- $f''(x)$ is concave up ($f''(x) > 0$) on $(-2, 4)$
- $f''(x) = 0$ (possible IP) on $x = -2, 4$ — when concavity switches

	-1	-2	1	4	5
$f''(x)$	−	0	+	0	−
$f(x)$	C down		C up		C down
		IP		IP	

when is graph (−) and (+)

Infliction point

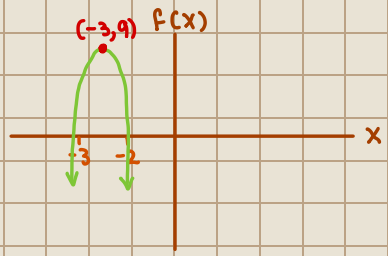
Find the degree

Since $f''(x)$ is degree 2, $f'(x)$ would be degree 3. Meaning $f(x)$ will be degree 4.

Example

Sketch a continuous function for each set of conditions.

a) $f''(x) = -2$ for all x, $f'(-3) = 0$, $f(-3) = 0$

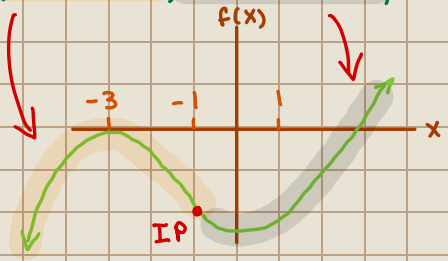


$f''(x)$ is Cdown when x is less than (-1)

$f''(x)$ is Cup when x is greater than (-1)

critical pts

b) $f''(x) < 0$ when $x < -1$, $f''(x) > 0$ when $x > -1$, $f(-3) = 0$, $f(-1) = 0$.



LESSON 2 - Laws of Log

$A, B, x, y > 0$, $B \neq 1$

Product Rule	$\log_B xy = \log_B x + \log_B y$
Quotient Rule	$\log_B\left(\frac{x}{y}\right) = \log_B x - \log_B y$
Power Rule	$\log_B x^r = r\log_B x$
Change of Base	$\log_B A = \frac{\log A}{\log B}$

① $\log_B 1 = 0$

② $\log_B B^x = x$

③ $B^{\log_B x} = x$

④ $y = \ln x \longleftrightarrow y = \log_e x$ (natural log)

⑤ $y = \log x \longleftrightarrow y = \log_{10} x$ (common log)

Example

1. $\ln e^x$ — $\ln x = \log_e x$ — x

$\log_e e^x$ — log rule ($\log_B B^x = x$)

$= x$

2. $5x \ln e^{2x}$

$= 5x \cdot 2x$

$= 10x^2$

(multiply w/ exponant

or $\ln e^{2x \cdot 5x}$

$= 2x \cdot 5x$

$= 10x^2$

3. $\ln 12 + \ln 5x - \ln 4x$

$= \ln\left(\frac{12 \cdot 5x}{4x}\right)$

$= \ln 15$

4. $2\ln 9 - \ln 3x$

$= 1\ln 9^2 - \ln 3x$

$= \ln\left(\frac{81}{3x}\right) = \ln\left(\frac{27}{x}\right)$

5. $\ln\sqrt{e}$

$= \ln e^{1/2}$

$= 1/2$

6. $\ln e \cdot \ln \sqrt[3]{e^5}$

$= \ln e \cdot \ln e^{5/3}$

$= 1 \cdot 5/3$

$= \frac{5}{3}$

7. $\ln(-2)$

= error

$\ln e$, $x > 0$

5. $e^x = 19$

$\ln e^x \quad \ln 19$

$x = \ln 19$

$x \doteq 2.94$

6. $\frac{2e^{3x}}{2} = \frac{571}{2}$

$e^{3x} = \frac{571}{2}$

$\ln e^{3x} = \ln\left(\frac{571}{2}\right)$

$\frac{3x}{3} = \frac{\ln\left(\frac{571}{2}\right)}{3}$

$x = \frac{\ln\left(\frac{571}{2}\right)}{3}$

$x \doteq 1.885$